

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12383059
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Chen; Ken-Ching et al.

Slide rail assembly

Abstract

A slide rail assembly includes a first rail, a second rail, a controlling member, an aiding member, a working member and a resilient structure. When the second rail is pulled from a retracted position relative to the first rail, the controlling member and the working member abut against each other to generate an acting force for driving the working member to rotate from a first state to a second state, such that a controlling portion of the controlling member is moved from a locking position relative to a predetermined path structure of the aiding member for allowing the second rail to be opened. The working member returns to the first state from the second state in response to a returning resilient force provided by the resilient structure.

Inventors: Chen; Ken-Ching (Kaohsiung, TW), Liang; Hsiu-Chiang (Kaohsiung, TW), Wang; Chun-Chiang (Kaohsiung, TW)

Applicant: KING SLIDE WORKS CO., LTD. (Kaohsiung, TW); KING SLIDE TECHNOLOGY CO., LTD. (Kaohsiung, TW)

Family ID: 88600253

Assignee: KING SLIDE WORKS CO., LTD. (Kaohsiung, TW); KING SLIDE TECHNOLOGY CO., LTD. (Kaohsiung, TW)

Appl. No.: 18/375991

Filed: October 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240360870 A1	Oct. 31, 2024

Foreign Application Priority Data

TW	112115890	Apr. 26, 2023
----	-----------	---------------

Publication Classification

Int. Cl.: F16C29/02 (20060101); A47B88/463 (20170101)

U.S. Cl.:

CPC A47B88/463 (20170101);

Field of Classification Search

CPC: A47B (88/463); A47B (2210/0008); A47B (2210/001); A47B (2210/0013)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8857868	12/2013	Liang	N/A	N/A
2004/0000851	12/2003	Lam Harn	312/334.7	A47B 88/447
2011/0129172	12/2010	Liang	384/22	A47B 88/463

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102755015	12/2011	CN	N/A
2272400	12/2018	EP	N/A
2008296616	12/2007	JP	N/A

Primary Examiner: Ing; Matthew W

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a slide rail product, and more specifically, to a slide rail assembly having a push-open mechanism with a protective function.

2. Description of the Prior Art

(2) With advancement of the industrial technology, there are various slide rail products with various functions in the market. For example, some of the slide rail products are slide rail assemblies with push-open mechanisms. When a drawer is mounted on a cabinet by a slide rail assembly with a push-open mechanism and is located at a retracted position relative to the cabinet, there is gap formed between a front panel portion of the drawer and the cabinet for allowing the push-open mechanism to drive the drawer to be opened relative to the cabinet by pushing the drawer relative to the cabinet.

(3) If the drawer is accidentally pulled from the retracted position due to the user's habit or the user's unawareness of the push-open mechanism, a movement of a movable rail of the slide rail assembly attached on the drawer may cause a structural damage and/or a functional failure of the push-open mechanism. Therefore, the push-open mechanism requires a protective function.

(4) For example, in U.S. Pat. No. 8,857,868 B2, it discloses a closing device including an engaging

mechanism. The engaging mechanism includes an engaging block, a positioning hook and a pin. The engaging block has an abutting surface and a guiding surface extending from the abutting surface. The positioning hook is connected to the engaging block and has a resilient portion and a stop which extends from the resilient portion. The stop is located adjacent to the abutting surface and away from the guiding surface. The pin is located at a position corresponding to the engaging block and slidably abuts against the abutting surface and the stop of the positioning hook. When a movable rail or a drawer is pulled to be opened, the pin abuts against the stop and the abutting surface, such that the resilient portion is resiliently deformed for allowing the pin to move over the stop to disengage from the abutting surface, which prevents abrasion or damage of the engaging hook, so as to achieve a purpose of protection.

(5) Furthermore, in EP 2272400 B1, it discloses a locking device adapted for a drawer slide. As shown in FIG. 5A of EP 2272400B1, the locking device includes a disc having a protruding portion configured to receive or lock an end of a switching element when the disc is located at an initial position. As shown in FIG. 5B of EP 2272400B1, when a movable rail or a drawer is pulled to be opened, the end of the switching element disengages from the protruding portion of the disc by a rotating movement of the disc, which achieves a purpose of protection. Besides, the disc can return to the initial position by a spring after the movable rail or the drawer is retracted.

(6) However, in order to meet various requirements, it becomes an important topic to provide an improved slide rail assembly.

SUMMARY OF THE INVENTION

(7) It is an objective of the present invention to provide a slide rail assembly having a push-open mechanism with a protective function.

(8) According to an aspect of the present invention, a slide rail assembly includes a first rail, a second rail, a driving device, a controlling member, an aiding member, a working member and a resilient structure. The second rail is displaceable relative to the first rail along a first direction or a second direction opposite to the first direction. The driving device includes a resilient member configured to provide a resilient force along the second direction. The controlling member is movably mounted on one of the first rail and the second rail. The controlling member includes a controlling portion. The aiding member is arranged on another one of the first rail and the second rail. The aiding member includes a predetermined path structure. The working member is movable relative to the aiding member. The working member includes a working portion located adjacent to the predetermined path structure. The resilient structure is configured to provide a returning resilient force to the working member for resiliently retaining the working member in a first state. When the second rail is located at a retracted position relative to the first rail, the controlling member and the working portion of the working member in the first state block each other, so as to position the controlling portion of the controlling member at a locking position for maintaining the resilient force provided by the resilient member. When the second rail is pushed by a first force from the retracted position relative to the first rail along the first direction, the aiding member guides the controlling portion of the controlling member to move from the locking position relative to the predetermined path structure, so as to allow the resilient member to drive the second rail to be opened along the second direction by the resilient force. When the second rail is pulled by a second force from the retracted position relative to the first rail along the second direction, the controlling member and the working portion of the working member in the first state abut against each other to generate an acting force for driving the working member to rotate from the first state to a second state, such that the controlling portion of the controlling member is moved from the locking position relative to the predetermined path structure for allowing the second rail to be opened along the second direction. The working member returns to the first state from the second state in response to the returning resilient force provided by the resilient structure.

(9) According to another aspect of the present invention, a slide rail assembly includes a first rail, a second rail, a driving device, a controlling member, an aiding member, a working member and a

resilient structure. The second rail is displaceable relative to the first rail along a first direction or a second direction opposite to the first direction. The driving device includes a resilient member configured to provide a resilient force along the second direction. The controlling member is movably mounted on one of the first rail and the second rail. The controlling member includes a controlling portion. The aiding member is arranged on another one of the first rail and the second rail. The aiding member includes a predetermined path structure. The predetermined path structure includes an entry section and a releasing section communicated with the entry section. The working member is movable relative to the aiding member. The working member includes a working portion located between the entry section and the releasing section. The working portion of the working member includes an abutting section. An offset angle is formed between an extending line passing through the abutting section and a center of the working member and an extending line passing through the center of the working member along the first direction or the second direction. The resilient structure is configured to provide a returning resilient force to the working member for resiliently retaining the working member in a first state. When the second rail is located at a retracted position relative to the first rail, the controlling member and the working portion of the working member in the first state block each other, so as to position the controlling portion of the controlling member at a locking position for maintaining the resilient force provided by the resilient member by a blocking of the controlling portion of the controlling member and the abutting section. When the second rail is pulled from the retracted position relative to the first rail along the second direction, the controlling member and the working portion of the working member in the first state abut against each other to generate an acting force for driving the working member to rotate from the first state to a second state, such that the controlling portion of the controlling member is moved from the locking position relative to the predetermined path structure for allowing the second rail to be opened along the second direction.

(10) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a slide rail assembly according to an embodiment of the present invention.
- (2) FIG. 2 is an exploded diagram of the slide rail assembly according to the embodiment of the present invention.
- (3) FIG. 3 is a partial exploded diagram of the slide rail assembly according to the embodiment of the present invention.
- (4) FIG. 4 and FIG. 5 are partial diagrams of the slide rail assembly at different views according to the embodiment of the present invention.
- (5) FIG. 6 is a diagram of the slide rail assembly as a second rail is located at a retracted position relative to a first rail according to the embodiment of the present invention.
- (6) FIG. 7 is a diagram of the slide rail assembly as the second rail is pushed from the retracted position relative to the first rail along a first direction according to the embodiment of the present invention.
- (7) FIG. 8 is a diagram of the slide rail assembly as the second rail is driven to displace relative to the first rail along a second direction by a driving device according to the embodiment of the present invention.
- (8) FIG. 9 is a diagram of the slide rail assembly as the second rail is located at an extended position relative to the first rail according to the embodiment of the present invention.

(9) FIG. 10 is a diagram of the slide rail assembly as the second rail is pulled from the retracted position relative to the first rail along the second direction according to the embodiment of the present invention.

DETAILED DESCRIPTION

(10) In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as “top”, “bottom”, “left”, “right”, “front”, “back”, etc., is used with reference to the orientation of the Figure(s) being described. The members of the present invention can be positioned in a number of different orientations. As such, the directional terminology is used for purposes of illustration and is in no way limiting. Accordingly, the drawings and descriptions will be regarded as illustrative in nature and not as restrictive. Also, if not specified, the term “connect” is intended to mean either an indirect or direct mechanical connection. Thus, if a first device is connected to a second device, that connection may be through a direct mechanical connection, or through an indirect mechanical connection via other devices and connections.

(11) As shown in FIG. 1 and FIG. 2, a slide rail assembly **20** includes a first rail **22**, a second rail **24**, a driving device **26**, a controlling member **28**, an aiding member **30**, a working member **32** and a resilient structure **34**.

(12) The second rail **24** is displaceable relative to the first rail **22** longitudinally. Preferably, the slide rail assembly **20** can further include a third rail **36** movably mounted between the first rail **22** and the second rail **24** and for extending a displacement of the second rail **24** relative to the first rail **22**. In practical, the first rail **22** can be mounted or fixed on a cabinet, and the second rail **24** can be configured to support a drawer. Understandably, in another embodiment, the third rail can be omitted.

(13) As shown in FIG. 1, the driving device **26** is configured to drive the second rail **24** to be opened after the second rail **24** is pushed from a retracted position R relative to the first rail **22** along a first direction D1, e.g., a retracting direction. The driving device **26** includes a resilient member **38**. In this embodiment, by way of example, the resilient member **38** can be a spring. However, the present invention is not limited to this embodiment. The resilient member **38** includes a first end **40a** and a second end **40b** opposite to the first end **40a**. The first end **40a** of the resilient member **38** is fixed on the first rail **22** by a connecting member **42**. A driving member **44** is arranged on the second end **40b** of the resilient member **38**. Preferably, a corresponding member **46** or a corresponding portion is arranged on the second rail **24** or the third rail **36** and located at a position corresponding to the driving member **44**, so as to cooperate with the driving member **44**. In this embodiment, by way of example, the corresponding member **46** can be arranged on the third rail **36**. However, the present invention is not limited to this embodiment. For example, in another embodiment, the corresponding member or the corresponding portion can be arranged on the second rail.

(14) The controlling member **28** is movably mounted on one of the first rail **22** and the second rail **24**. In this embodiment, by way of example, the controlling member **28** can be movably mounted on the first rail **22**. Specifically, the controlling member **28** can be pivotally connected to the first rail **22** by a shaft **48**, such that the controlling member **28** is pivotable relative to the first rail **22**. Preferably, as shown in FIG. 6, the first rail **22** includes a first restraining feature **50**, and the controlling member **28** includes a second restraining feature **51**, wherein one of the first restraining feature **50** and the second restraining feature **51** is an opening or a recess portion, and the other one of the first restraining feature **50** and the second restraining feature **51** is an extending leg stretching into the opening or the recess portion and configured to abut against a wall of the opening or the recess portion, such that the controlling member **28** is only allowed to pivot relative to the first rail **22** within a limited range. In this embodiment, by way of example, the first restraining feature **50** can be the opening, and the second restraining feature **51** can be the extending leg. However, the

present invention is not limited to this embodiment.

(15) The controlling member **28** includes a controlling portion **52**. In this embodiment, by way of example, the controlling portion **52** can be a protruding portion or a pin. However, the present invention is not limited to this embodiment.

(16) Preferably, the first rail **22** includes a front end portion **54a** and a rear end portion **54b**, and the controlling member **28** is located adjacent to the front end portion **54a** of the first rail **22**.

(17) As shown in FIG. 3, FIG. 4 and FIG. 5, the aiding member **30** is arranged on the other one of the first rail **22** and the second rail **24** where the controlling member **28** is not disposed. In this embodiment, by way of example, the aiding member **30** can be arranged on the second rail **24**. Preferably, the aiding member **30** can be connected to the second rail **24** directly or indirectly. The second rail **24** includes a front end portion **56a** and a rear end portion **56b**, and the aiding member **30** is arranged on the second rail **24** and located adjacent to the front end portion **56a** of the second rail **24**.

(18) The working member **32** is movable relative to the aiding member **30**. In this embodiment, by way of example, the working member **32** can be movably mounted on the aiding member **30**.

(19) The resilient structure **34** is configured to provide a returning resilient force K to the working member **32** for resiliently retaining the working member **32** in a first state S1 as shown in FIG. 4. In this embodiment, by way of example, the resilient structure **34** can be a leaf spring. However, the present invention is not limited to this embodiment. For example, in another embodiment, the resilient structure can be a spring or any other resiliently deformable object, e.g., a rubber cylinder.

(20) Preferably, the aiding member **30** is configured to provide a moving space M for the working member **32**. As shown in FIG. 3, the moving space M penetrates through a first portion **58** and a second portion **60** of the aiding member **30**.

(21) The working member **32** includes a working portion **66**. Preferably, the working member **32** further includes a main body portion **62** and an extending section **64**. The extending section **64** extends from the main body portion **62**. The working portion **66** protrudes from a surface **67** of the main body portion **62**. As shown in FIG. 4, the main body portion **62** and the extending section **64** are located adjacent to the first portion **58** of the aiding member **30**. As shown in FIG. 5, the working portion **66** passes through the moving space M and is located adjacent to the second portion **60** of the aiding member **30**.

(22) Preferably, the aiding member **30** is further configured to provide an aiding receiving space Q for receiving the resilient structure **34**. The aiding receiving space Q is located adjacent to the moving space M. The resilient structure **34** includes a first predetermined portion **68** and a second predetermined portion **70**. As shown in FIG. 3 and FIG. 4, the first predetermined portion **68** is configured to abut against a wall **72** of the aiding member **30**, and the second predetermined portion **70** is configured to abut against the extending section **64** of the working member **32**.

(23) As shown in FIG. 5, the aiding member **30** includes a predetermined path structure **74**, and the working portion **66** of the working member **32** is located adjacent to the predetermined path structure **74**.

(24) Preferably, as shown in FIG. 5, the working portion **66** of the working member **32** is a heart-shaped protrusion and surrounded by the predetermined path structure **74**. The predetermined path structure **74** includes an entry section **76a** and a releasing section **76b** communicated with the entry section **76a**. The working portion **66** of the working member **32** is located between the entry section **76a** and the releasing section **76b**.

(25) Preferably, the aiding member **30** further includes a guiding section **78**. In this embodiment, by way of example, the guiding section **78** can be an inclined surface or an arc surface. The working portion **66** of the working member **32** further includes an abutting section **80**.

(26) Preferably, as shown in FIG. 3 or FIG. 5, the surface **67** of the main body portion **62** of the working member **32**, is formed in a substantially circular shape.

(27) As shown in FIG. 6, when the second rail **24** is located at the retracted position R relative to

the first rail **22**, the controlling member **28** and the working portion **66** of the working member **32** in the first state **S1** can block each other, so as to position the controlling portion **52** of the controlling member **28** at a locking position **L** for maintaining a resilient force **J** provided by the resilient member **38** along a second direction **D2**, e.g., an opening direction, wherein the second direction **D2** is opposite to the first direction **D1**.

(28) Preferably, when the second rail **24** is located at the retracted position **R** relative to the first rail **22**, the controlling portion **52** of the controlling member **28** and the abutting section **80** of the working portion **66** of the working member **32** in the first state **S1** block each other, such that the controlling member **28** and the working portion **66** of the working member **32** in the first state **S1** block each other, so as to position the controlling portion **52** of the controlling member **28** at the locking position **L** for maintaining the resilient force **J** provided by the resilient member **38** along the second direction **D2** by a blocking of the controlling portion **52** of the controlling member **28** and the abutting section **80** of the working portion **66** of the working member **32**. Furthermore, a locking space **X** is defined between the guiding section **78** and the abutting section **80** and configured to receive the controlling portion **52** of the controlling member **28**.

(29) Preferably, an offset angle **A** is formed between an extending line passing through the abutting section **80** and a center **82** of the working member **32**, e.g., a center of the main body portion **62** of the working member **32**, and an extending line passing through the center **82** of the working member **32** along the first direction **D1** or the second direction **D2**.

(30) As shown in FIG. 7, FIG. 8 and FIG. 9, when the second rail **24** is pushed by a first force **F1** from the retracted position **R** relative to the first rail **22** along the first direction **D1**, the aiding member **30** guides the controlling portion **52** of the controlling member **28** to move from the locking position **L** relative to the predetermined path structure **74**, so as to allow the resilient member **38** to drive the second rail **24** to be opened relative to the first rail **22** along the second direction **D2** by the resilient force **J**, e.g., until the second rail **24** is displaced to an extended position **E**. In other words, the slide rail assembly **20** is configured to provide a push-open mechanism which enables the second rail **24** to be opened relative to the first rail **22** along the second direction **D2** by pushing the second rail **24** from the retracted position **R** along the first direction **D1**.

(31) Preferably, as shown in FIG. 7, when the second rail **24** is pushed by the first force **F1** from the retracted position **R** relative to the first rail **22** along the first direction **D1**, the controlling portion **52** of the controlling member **28** and the guiding section **78** of the aiding member **30** abut against each other, such that the controlling portion **52** of the controlling member **28** is guided to move from the locking position **L** into the releasing section **76b** of the predetermined path structure **74** by the guiding section **78**, so as to allow the resilient member **38** to drive the second rail **24** to be opened along the second direction **D2**. For example, as shown in FIG. 8 and FIG. 9, the second end **40b** of the resilient member **38** can push the corresponding member **46** on the third rail **36** by the driving member **44** to drive the third rail **36** and the second rail **24** to be opened along the second direction **D2**.

(32) Preferably, the aiding member **30** further includes a guiding feature **84**. In this embodiment, by way of example, the guiding feature **84** can be an inclined surface or an arc surface. As shown in FIG. 8 and FIG. 9, during a displacement of the second rail **24** along the second direction **D2** in response to the resilient force **J** provided by the resilient member **38**, the aiding member **30** drives the controlling portion **52** of the controlling member **28** to move out of the predetermined path structure **74** to a position corresponding to the entry section **76a** of the aiding member **30** as shown in FIG. 8 and FIG. 9 by an abutment of the guiding feature **84** and the controlling portion **52** of the controlling member **28**. Therefore, during a displacement of the second rail **24** from the extended position **E** to the retracted position **R** along the first direction **D1**, the controlling portion **52** of the controlling member **28** can return back to the locking position **L** as shown in FIG. 6 by entering into the predetermined path structure **74** through the entry section **76a**.

(33) As shown in FIG. 6 and FIG. 10, when the second rail 24 is pulled by a second force F2 from the retracted position R relative to the first rail 22 along the second direction D2, the controlling portion 52 of the controlling member 28 and the working portion 66 of the working member 32 can abut against each other to generate an acting force V for driving the working member 32 to rotate from the first state S1 to a second state S2 as shown in FIG. 10, such that the controlling portion 52 of the controlling member 28 is moved from the locking position L relative to the predetermined path structure 74 for allowing the second rail 24 to be opened along the second direction D2.

(34) Preferably, when the second rail 24 is pulled by the second force F2 from the retracted position R relative to the first rail 22 along the second direction D2, the controlling portion 52 of the controlling member 28 and the working portion 66 of the working member 32 can abut against each other to generate the acting force V for driving the working member 32 to rotate from the first state S1 as shown in FIG. 6 to the second state S2 as shown in FIG. 10 easily due to the offset angle A, such that the controlling portion 52 of the controlling member 28 can be driven by the working portion 66 of the working member 32 to move from the locking position L relative to the predetermined path structure 74 for allowing the second rail 24 to be opened along the second direction D2 easily. Besides, the working portion 66 of the working member 32 further includes a guiding surface 55 located adjacent to the abutting section 80. The controlling portion 52 of the controlling member 28 includes a guiding contour 59. In this embodiment, by way of example, the guiding surface 55 can be an inclined surface or an arc surface, and the guiding contour 59 can be formed in a circular shape or an arc shape. When the second rail 24 is pulled by the second force F2 from the retracted position R relative to the first rail 22 along the second direction D2, the controlling portion 52 of the controlling member 28 and the working portion 66 of the working member 32 can abut against each other to generate the acting force V by a cooperation of the guiding surface 55 and the guiding contour 59, which facilitates not only a rotating movement of the working member 32 from the first state S1 as shown in FIG. 6 to the second state S2 as shown in FIG. 10 but also a movement of the controlling portion 52 of the controlling member 28 from the locking position L into the releasing portion 76b of the predetermined path structure 74.

(35) Furthermore, as shown in FIG. 10, when the working member 32 is in the second state S2, the resilient structure 34 is resiliently deformed. Therefore, once the second rail 24 is displaced from the retracted position R to a predetermined position, e.g., the extended position E, relative to the first rail 22 along the second direction D2, the working member 32 can return to the first state S1 as shown in FIG. 9 from the second state S2 as shown in FIG. 10 in response to the returning resilient force K provided by the resilient structure 34.

(36) From the above, when the second rail 24 is located at the retracted position R relative to the first rail 22, the second rail 24 can be opened normally, i.e., by pushing the second rail 24 in the first direction D1. However, if the second rail 24 is pulled in the second direction D2 inappropriately, the rotating movement of the working member 32 relative to the aiding member 30 from the first state S1 to the second state S2, allows the controlling portion 52 of the controlling member 28 to move from the locking position L into the releasing section 76b of the predetermined path structure 74, which prevents abrasion or damage of the working portion 66 of the working member 32 and/or the controlling portion 52 of the controlling member 28 due to friction or a collision, so as to achieve a purpose of protection.

(37) In summary, the slide rail assembly 20 includes the following characteristics.

(38) 1. When the second rail 24 is pulled by the second force F2 from the retracted position R relative to the first rail 22 along the second direction D2, the controlling portion 52 of the controlling member 28 and the working portion 66 of the working member 32 can abut against each other to generate the acting force V for driving the working member 32 to move from the first state S1 to the second state S2, such that the controlling portion 52 of the controlling member 28 is moved from the locking position L into the releasing section 76b of the predetermined path structure 74 for allowing the second rail 24 to be opened along the second direction D2. The

rotating movement of the working member 32 allows the controlling portion 52 of the controlling member 28 to move from the locking position L, which prevents abrasion or damage of the working portion 66 of the working member 32 and/or the controlling portion 52 of the controlling member 28 due to friction or collision.

(39) 2. The working member 32 can return to the first state S1 as shown in FIG. 9 from the second state S2 as shown in FIG. 10 in response to the returning resilient force K provided by the resilient structure 34 once the second rail 24 is displaced from the retracted position R to the predetermined position, e.g., the extended position E.

(40) 3. When the second rail 24 is pulled by the second force F2 from the retracted position R relative to the first rail 22 along the second direction D2, the controlling portion 52 of the controlling member 28 and the working portion 66 of the working member 32 can abut against each other to generate the acting force V for driving the working member 32 to rotate from the first state S1 as shown in FIG. 6 to the second state S2 as shown in FIG. 10 easily due to the offset angle A, such that the controlling portion 52 of the controlling member 28 can be driven by the working portion 66 of the working member 32 to move from the locking position L relative to the predetermined path structure 74 for allowing the second rail 24 to be opened along the second direction D2 easily.

(41) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A slide rail assembly comprising: a first rail; a second rail displaceable relative to the first rail along a first direction or a second direction opposite to the first direction; a driving device comprising a resilient member configured to provide a resilient force along the second direction; a controlling member movably mounted on one of the first rail and the second rail, the controlling member comprising a controlling portion; an aiding member arranged on another one of the first rail and the second rail, the aiding member comprising a predetermined path structure; a working member movable relative to the aiding member, the working member comprising a working portion located adjacent to the predetermined path structure; and a resilient structure configured to provide a returning resilient force to the working member for resiliently retaining the working member in a first state; wherein when the second rail is located at a retracted position relative to the first rail, the controlling member and the working portion of the working member in the first state block each other, so as to position the controlling portion of the controlling member at a locking position for maintaining the resilient force provided by the resilient member; wherein when the second rail is pushed by a first force from the retracted position relative to the first rail along the first direction, the aiding member guides the controlling portion of the controlling member to move from the locking position relative to the predetermined path structure, so as to allow the resilient member to drive the second rail to be opened along the second direction by the resilient force; wherein when the second rail is pulled by a second force from the retracted position relative to the first rail along the second direction, the controlling member and the working portion of the working member in the first state abut against each other to generate an acting force for driving the working member to rotate from the first state to a second state, such that the controlling portion of the controlling member is moved from the locking position relative to the predetermined path structure for allowing the second rail to be opened along the second direction; wherein the working member returns to the first state from the second state in response to the returning resilient force provided by the resilient structure; wherein the aiding member is configured to provide a moving space for the working member, the moving space penetrates through a first portion and a second portion of

the aiding member, the working member further comprises a main body portion and an extending section, the extending section extends from the main body portion, the working portion protrudes from a surface of the main body portion, the main body portion and the extending section are located adjacent to the first portion of the aiding member, and the working portion passes through the moving space and is located adjacent to the second portion of the aiding member.

2. The slide rail assembly of claim 1, wherein the controlling member is pivotally connected to the first rail by a shaft.

3. The slide rail assembly of claim 1, further comprising a third rail movably mounted between the first rail and the second rail, and the third rail being configured to extend a displacement of the second rail relative to the first rail.

4. The slide rail assembly of claim 1, wherein the aiding member is arranged on the second rail, the predetermined path structure comprises an entry section and a releasing section communicated with the entry section, the working member is movably mounted on the aiding member, and the working portion of the working member is located between the entry section and the releasing section.

5. The slide rail assembly of claim 4, wherein the aiding member further comprises a guiding section, when the second rail is pushed by the first force from the retracted position relative to the first rail along the first direction, the controlling portion of the controlling member is guided to move from the locking position into the releasing section of the predetermined path structure by the guiding section, so as to allow the resilient member to drive the second rail to be opened along the second direction.

6. The slide rail assembly of claim 5, wherein the working portion of the working member comprises an abutting section, an offset angle is formed between an extending line passing through the abutting section and a center of the working member and an extending line passing through the center of the working member along the first direction or the second direction, and the controlling portion of the controlling member abuts against the abutting section when the controlling portion of the controlling member is located at the locking position.

7. The slide rail assembly of claim 6, wherein when the second rail is pulled by the second force from the retracted position relative to the first rail along the second direction, the controlling portion of the controlling member and the working portion of the working member in the first state abut against each other to generate the acting force for driving the working member to rotate from the first state to the second state due to the offset angle, such that the controlling portion of the controlling member is moved from the locking position into the releasing section of the predetermined path structure for allowing the second rail to be opened along the second direction.

8. A slide rail assembly comprising: a first rail; a second rail displaceable relative to the first rail along a first direction or a second direction opposite to the first direction; a driving device comprising a resilient member configured to provide a resilient force along the second direction; a controlling member movably mounted on one of the first rail and the second rail, the controlling member comprising a controlling portion; an aiding member arranged on another one of the first rail and the second rail, the aiding member comprising a predetermined path structure, the predetermined path structure comprising an entry section and a releasing section communicated with the entry section; a working member movable relative to the aiding member, the working member comprising a working portion located between the entry section and the releasing section, the working portion of the working member comprising an abutting section, an offset angle being formed between an extending line passing through the abutting section and a center of the working member and an extending line passing through the center of the working member along the first direction or the second direction; and a resilient structure configured to provide a returning resilient force to the working member for resiliently retaining the working member in a first state; wherein when the second rail is located at a retracted position relative to the first rail, the controlling member and the working portion of the working member in the first state block each other, so as to position the controlling portion of the controlling member at a locking position for maintaining the

resilient force provided by the resilient member by a blocking of the controlling portion of the controlling member and the abutting section; wherein when the second rail is pulled from the retracted position relative to the first rail along the second direction, the controlling member and the working portion of the working member in the first state abut against each other to generate an acting force for driving the working member to rotate from the first state to a second state, such that the controlling portion of the controlling member is moved from the locking position relative to the predetermined path structure for allowing the second rail to be opened along the second direction; wherein the aiding member is configured to provide a moving space for the working member, the moving space penetrates through a first portion and a second portion of the aiding member, the working member further comprises a main body portion and an extending section, the extending section extends from the main body portion, the working portion protrudes from a surface of the main body portion, the main body portion and the extending section are located adjacent to the first portion of the aiding member, and the working portion passes through the moving space and is located adjacent to the second portion of the aiding member.

9. The slide rail assembly of claim 8, wherein when the second rail is pulled from the retracted position relative to the first rail along the second direction, the controlling portion of the controlling member and the working portion of the working member in the first state abut against each other to generate the acting force for driving the working member to rotate from the first state to the second state due to the offset angle, such that the controlling portion of the controlling member is moved from the locking position into the releasing section of the predetermined path structure for allowing the second rail to be opened along the second direction.

10. The slide rail assembly of claim 8, wherein the controlling member is pivotally connected to the first rail by a shaft.

11. The slide rail assembly of claim 8, further comprising a third rail movably mounted between the first rail and the second rail, and the third rail being configured to extend a displacement of the second rail relative to the first rail.

12. The slide rail assembly of claim 8, wherein the aiding member is arranged on the second rail, and the working member is movably mounted on the aiding member.

13. The slide rail assembly of claim 8, wherein the working member returns to the first state from the second state in response to the returning resilient force provided by the resilient structure.

14. The slide rail assembly of claim 8, wherein the aiding member further comprises a guiding section, when the second rail is pushed from the retracted position relative to the first rail along the first direction, the controlling portion of the controlling member is guided to move from the locking position into the releasing section of the predetermined path structure by the guiding section, so as to allow the resilient member to drive the second rail to be opened along the second direction.
